Estuarine, Coastal and Shelf Science 340 (2026) 110103

Contents lists available at ScienceDirect

Estuarine, Coastal and Shelf Science

journal homepage: www.elsevier.com/locate/ecss

Source-aware screening of long-term salinity trends in public estuarine and coastal monitoring records

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ARTICLE INFO

Keywords:
Estuarine salinity
Coastal monitoring
Water quality
Trend analysis
Quality control
Data synthesis

ABSTRACT

Long salinity records are central to understanding estuarine change, but they are scattered across monitoring programs that differ in station siting, sampling frequency, quality-control practice and data access. Here, a source-aware (i.e., provider-stratified and station-identity-preserving) estuarine and coastal salinity resource was assembled containing 95,008,350 harmonized records from 5383 source-aware station entries and 13 public data sources spanning 1963-2026. The study asks whether public monitoring archives can support reliable broad inference on estuarine salinity trends, or whether apparent trend evidence is dominated by monitoring design and source structure. All observations were retained with provenance and quality-control fields, then aggregated to station-month series for a descriptive trend screen. Under the primary eligibility rule of at least 10 years of record span with at least 60 quality-control-pass monthly means distributed across that span, 436 source-aware station series were trend eligible. Nominal Kendall/Sen estimates classified 99 series as increasing, 76 as decreasing and 261 as not significant. Fixed-station analyses excluding gridded CTD cells retained the same heterogeneous pattern, and direct-salinity-only sensitivity gave nearly identical counts. However, statistical inference was more restricted: false-discovery-rate control retained 127 significant trends, annual aggregation reduced nominal detections substantially, and 32 of 436 broad trend classes changed when flagged observations were included. The synthesis provides an auditable observational baseline for estuarine salinity comparison and identifies where future driver attribution and archive completion are most needed.

1. Introduction

Salinity is one of the organizing variables of estuarine systems. It governs stratification, habitat boundaries, biogeochemical cycling rates, freshwater exposure, drinking-water suitability, and the interpretation of co-located water-quality parameters including dissolved oxygen, pH, nutrients, and turbidity (IPCC, 2022; Kaushal et al., 2018; Costa et al., 2023). Estuaries and coastal wetlands support fisheries, nutrient filtration, storm-surge attenuation, and carbon sequestration services whose ecological value depends on salinity remaining within ranges tolerated by resident communities. Long salinity records can reveal salt intrusion, freshening, altered residence time, and changes in estuarine exchange, but the records needed for broad comparison are rarely held in a single archive. They are produced by agencies and research programs with different mandates, sampling designs, quality flags, station identifiers, and reporting conventions (National Estuarine Research Reserve System Centralized Data Management Office, 2026; NOAA Center for Operational Oceanographic Products and Services, 2026; U.S. Geological Survey, 2026).

This fragmentation creates a practical problem for synthesis. A data source with millions of observations may represent a small number of high-frequency stations, whereas a survey archive may contain many station names with short or discontinuous records. If these differences are ignored, a large compilation can look more spatially representative than it is. For estuarine and coastal salinity, this is especially important because station siting and monitoring design are part of the environmental signal: a harbor tide gauge, a tidal marsh sensor, a compliance grab-sample station, and a gridded coastal CTD cell are different units of evidence. Public compliance monitoring networks (e.g., NOAA CO-OPS, UK Environment Agency, NSW EPA) and research-based programs (e.g., NERRS SWMP, Ifremer QUADRIGE/REPHY, CBP) differ in their station siting logic, sampling frequency, and continuity objectives, so records from these sources are not directly interchangeable in a pooled trend analysis. Moreover, most individual fixed-station records span only 10-30 years, which is comparable to the dominant periods of internal climate modes such as ENSO, the PDO, and the NAO; this means that many apparent trends may reflect decadal variability rather than long-term directional change, and that cross-archive synthesis is needed to

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https://doi.org/10.1016/j.ecss.2026.110103
Received 8 May 2026; Received in revised form 11 June 2026; Accepted 7 July 2026
Available online 7 July 2026
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assess whether patterns are consistent across independent programs. In harbour environments, wave-driven circulation and the presence of coastal structures such as wave screens alter local exchange flows and salinity exposure independently of basin-scale tidal and riverine forcing (Biondi et al., 2026a), so harbour monitoring records carry site-specific hydrodynamic context not shared by open-estuary sensors.

Several long-term monitoring programs have documented salinity changes at regional scales. NERRS SWMP multi-reserve analyses have shown spatially variable dissolved-oxygen and salinity dynamics across East Coast and Gulf of Mexico reserves (Wenner et al., 2004; Baumann & Smith, 2017). NOAA CO-OPS harbor-entrance records indicate positive salinity trends at several major ports consistent with sea-level rise and altered freshwater inputs. Ifremer QUADRIGE/REPHY stations along the French Atlantic coast show long-term increases in coastal salinity linked to reduced riverine freshwater input. The freshwater salinization syndrome documented by Kaushal et al. (2018) in North American streams shows that anthropogenic ion loading, road salt, and land-use change are widespread drivers of salinity increase independent of marine salt intrusion. Hydrodynamic model projections for European estuaries (Lee et al., 2024, 2025) and global tidal rivers (Li et al., 2025) project increasing salt intrusion risk under continued sea-level rise and altered discharge regimes. These results collectively motivate a coordinated observational baseline that can anchor projection validation and track whether projected trends are already detectable in monitoring archives.

Existing studies have documented important salinity changes in individual estuaries and broader freshwater salinization contexts (Kaushal et al., 2018; IPCC, 2022; Costa et al., 2023; Li et al., 2025; Beemster et al., 2026; Biondi et al., 2026b). Modeling studies have projected future increases in salt intrusion under warming and sea-level rise at European and broader scales (Lee et al., 2024, 2025): Lee et al. (2024) showed increasing risks of extreme salt intrusion events - defined as periods when salinity at the freshwater intake threshold is exceeded for at least 30 consecutive days under ensemble hydrodynamic projections - in European estuaries, while Lee et al. (2025) extended this framework to project salt intrusion increases under future environmental forcing. Those projections motivate a broader observational baseline, but they do not substitute for a careful accounting of what monitoring archives currently show. The central question addressed here is whether public monitoring archives can support reliable regional inference on estuarine salinity trends, or whether apparent trends are dominated by monitoring design, source structure, and record eligibility.

In this study, direct salinity observations and harmonized salinity equivalents were compiled from public estuarine and coastal monitoring sources into a source-aware salinity resource - that is, a compilation in which each record retains its provider prefix and station identity so that records from different programs are never silently merged. Three questions are addressed: (1) Can public monitoring archives, taken together, support reliable inference on broad estuarine salinity trends, or is apparent trend evidence dominated by monitoring design and source structure? (2) How do trend-eligible series and direction counts change under conservative sensitivity tests including multiple-testing control, annual aggregation, and autocorrelation correction? (3) Which regional signals in the compiled archives are coherent enough to motivate attribution-focused follow-up analyses? All observations were retained with quality-control annotations, and monthly source-aware station trend screens were computed with parallel sensitivity tests. The resulting contribution is an auditable observational baseline and an evaluation of the extent to which public monitoring archives can currently support estuarine salinity trend inference.

2. Materials and methods

2.1. Data sources and harmonization

Data assembly followed five sequential steps.

(a) Source identification and scoping. Thirteen public estuarine and coastal monitoring programs were selected on the basis of providing salinity or conductivity records with public access and documented station metadata. These are listed in Table 1 and mapped in Fig. 1.

(b) Source-specific ingestion and canonical table construction. Source-specific ingestion scripts produced a canonical parquet table with source identifiers, source-aware station identifiers (provider prefix plus original station code), timestamps, coordinates where available, harmonized salinity values, raw values, and provenance fields. Source-aware station identity was preserved throughout to avoid merging unrelated station codes from different providers.

(c) Unit conversion. Direct salinity records were retained as practical salinity values where available, referenced to the Practical Salinity Scale 1978 (PSS-78; UNESCO, 1981; Fofonoff and Millard, 1983). For sources reporting specific conductance or electrical conductivity (EC) rather than salinity, values were converted only when the source reported conductivity standardized to 25 degrees C or when the provider metadata described the field as specific conductance. The conversion used

$$S = EC\left(\mu S/cm\right)\cdot\left(\frac{35}{53087}\right) = 0.000659\cdot EC$$

on the PSS-78 scale (UNESCO, 1981). This is a linear approximation derived from the conductivity of standard seawater at S = 35, T = 25 degrees C; the full PSS-78 standard uses polynomial functions of conductivity ratio, temperature, and pressure (Fofonoff and Millard, 1983), and the linearization is least defensible in freshwaters with non-marine ionic composition (Kaushal et al., 2018). All conductivity-derived salinity equivalents were flagged with conversion provenance and source-specific caveats, and a direct-salinity-only sensitivity test was used to check whether excluding EC-derived values changed the trend conclusions.

Table 1
Source inventory and trend eligibility. Fixed-station eligible excludes Taiwan ODB gridded CTD cells. Direct-salinity eligible excludes EC-derived salinity equivalents from USGS NWIS and GEMStat sources.
Acronyms: CBP = Chesapeake Bay Program; DRBC = Delaware River Basin Commission; GEMStat = Global Environment Monitoring System for freshwater (UNEP/GEMS Water); Ifremer = Institut Francais de Recherche pour l'Exploitation de la Mer; NERRS = National Estuarine Research Reserve System; NOAA CO-OPS = NOAA Center for Operational Oceanographic Products and Services; NSW EPA = New South Wales Environment Protection Authority; PANGAEA = Publisher for Earth and Environmental Science; RWS = Rijkswaterstaat (Dutch national water authority); UK EA = United Kingdom Environment Agency; USGS NWIS = U.S. Geological Survey National Water Information System.

Source	Records	Entries	Eligible	Fixed eligible	Direct-salinity eligible
CBP	839,717	854	16	16	16
DRBC	4332	57	22	22	22
GEMStat Argentina	658	43	0	0	0
GEMStat Mexico	30,772	1004	0	0	0
Ifremer	109,191	504	147	147	147
NERRS	82,451,362	173	117	117	117
NOAA CO-OPS	9,494,695	10	5	5	5
NSW EPA	4248	177	12	12	12
PANGAEA	25,905	34	0	0	0
RWS	3664	3	3	3	3
Taiwan ODB grid	30,726	1706	111	0	111
UK EA	37,305	812	0	0	0
USGS NWIS	1,975,775	6	3	3	0
Total	95,008,350	5383	436	325	433

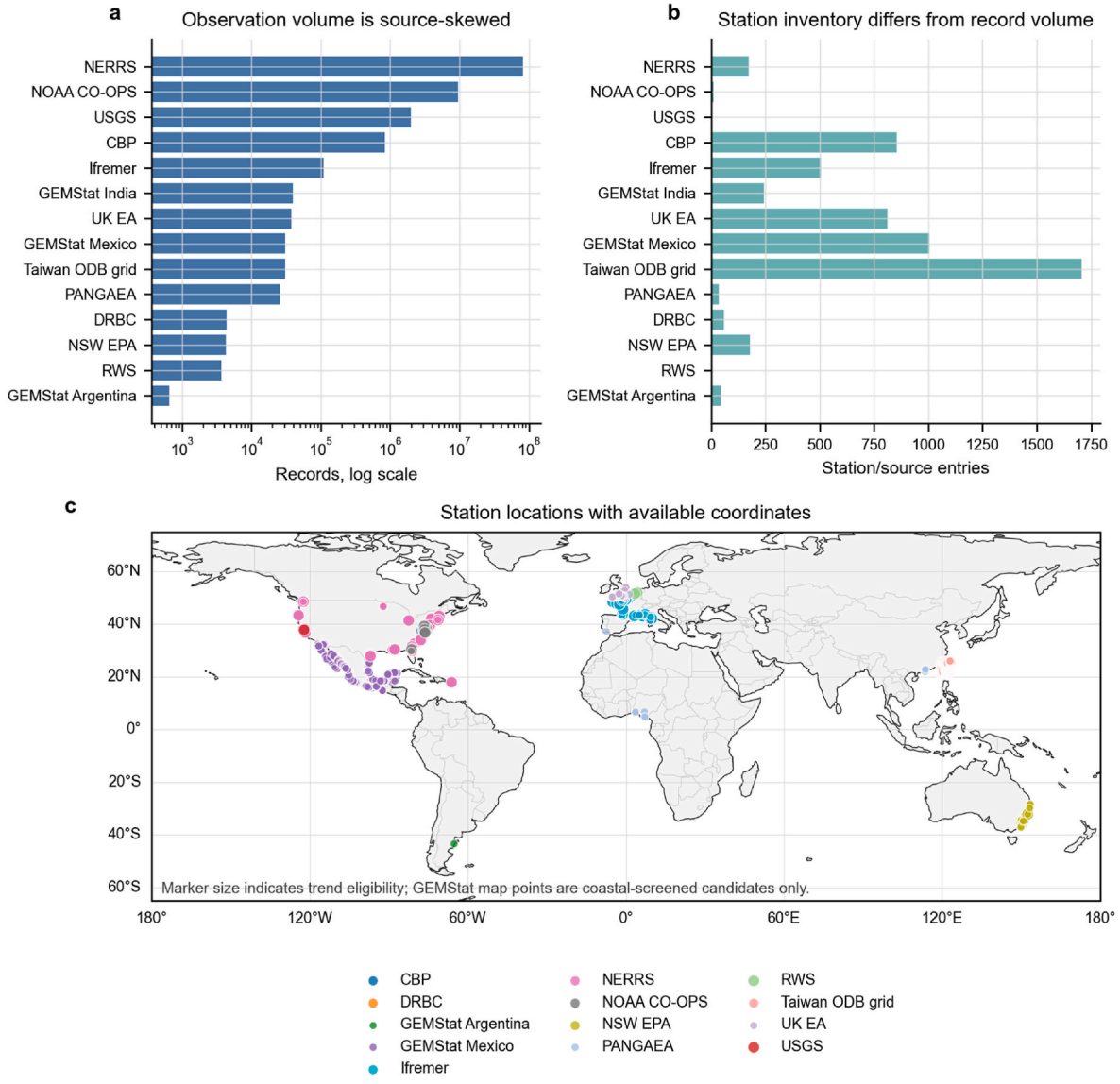


Fig. 1. Resource coverage and source imbalance. (a) Total observation volume by source (bar chart), showing the dominance of NERRS SWMP with 82,451,362 records (86.8% of all observations). (b) Station inventory by source (bar chart), showing the number of source-aware station entries per provider regardless of trend eligibility. (c) Coordinate-available station locations mapped globally; dot area is proportional to the number of trend-eligible source-aware station series at that location, with the minimum dot representing 1 eligible series and the maximum representing the largest single-location count. The map shows monitoring coverage only and does not imply representative sampling of all estuaries.

(d) Spatial-domain screening. Spatial-domain screening was applied separately from unit conversion. GEMStat (Global Environment Monitoring System for freshwater; UNEP/GEMS Water; GEMStat, 2026a; GEMStat, 2026b) records were retained as salinity-context records but excluded from the estuarine headline denominator when stations were more than 75 km from the Natural Earth 10 m coastline or were otherwise identifiable as inland salinization context from coordinates and metadata. The 75 km cutoff was used as a deliberately conservative first-pass buffer, wide enough to retain coastal-lagoon and tidal-river candidates despite coastline-resolution error, while requiring any more inland record to have explicit metadata support before contributing to estuarine trend inference. Exclusion reasons are listed in the reproducibility package.

(e) Station identity preservation. Source-aware station identity was used throughout the workflow to avoid merging unrelated codes from different providers. Table 1 and Fig. 1 show the resulting source inventory, station counts, and coordinate coverage.

2.2. Quality control

Quality control retained all observations and added flags for provider indicators, duplicate station timestamps, range checks, improbable jumps, flatline or stuck-sensor behavior, temporal gaps and metadata limitations. Specific thresholds applied in the primary screen were: practical salinity range 0-42; temporal jump greater than 15 practical-salinity units per hour between consecutive sub-hourly measurements; and flatline runs of 96 or more consecutive identical values, equivalent to four consecutive days at 15-min measurement frequency. These thresholds were chosen as conservative screening rules to flag values outside the expected open-estuary range, changes faster than plausible estuarine mixing at sub-hourly scale, and multi-day unchanged sensor values; they were not used to delete observations. The primary trend screen used observations marked as quality-control pass. Parallel all-observation trend fields were generated to test whether flagged observations changed broad trend classes.

2.3. Monthly aggregation and trend estimation

Records were aggregated to source-aware station-month means. A per-station monthly climatology was computed across the full station record and subtracted to produce seasonal anomaly series. Kendall tau was used as a Mann-Kendall-style descriptive trend screen and Theil-Sen/Sen slopes were reported in practical-salinity units per decade (Mann, 1945; Kendall, 1975; Sen, 1968). Nominal Kendall p values were retained as screening diagnostics, but inference was based on the hierarchy of false-discovery-rate control, annual aggregation and autocorrelation sensitivity rather than on unadjusted monthly p values alone. A series was eligible for the primary screen if it spanned at least 10 years and contained at least 60 observed monthly means distributed across the record span, so that eligibility required sustained temporal coverage rather than concentrated sampling within a short sub-period.

2.4. Robustness analyses

Eight sensitivity tests were applied to assess the robustness of the primary trend conclusions.

(1) Stricter eligibility threshold. The primary 10-year/60-month rule was tightened to 15 years and 120 observed monthly means, requiring longer and denser records. This tests whether the trend count distribution changes when marginal short-record series are excluded.
(2) Relaxed eligibility threshold. The threshold was loosened to 8 years and 48 monthly means to assess whether the direction pattern changes when more series with shorter coverage are included.
(3) Annual aggregation. Monthly station-month means were further aggregated to annual station means and the eligibility screen was re-applied at annual resolution (minimum 10 years with at least one annual mean each year). Annual aggregation reduces the influence of seasonal cycles and sub-annual autocorrelation on the Kendall test.
(4) Benjamini-Hochberg false-discovery-rate (FDR) control. Nominal monthly Kendall p values are not independent across the 436-series denominator. FDR control at $q < 0.05$ (Benjamini and Hochberg, 1995) was applied to identify the subset of series whose significance survives multiple-testing adjustment.
(5) Lag-one autocorrelation correction via trend-free pre-whitening (TFPW; Yue et al., 2002). In the TFPW approach, the Sen slope was first removed from each station series, the lag-one autoregressive coefficient was estimated from the detrended residuals, the residuals were pre-whitened by subtracting the lagged term, and the trend was then re-added before re-applying the Kendall test. TFPW is one of several pre-whitening approaches and has known limitations, including potential inflation of Type I error under positive autocorrelation (Zhang and Zwiers, 2004; Burger, 2017).
(6) Source leave-one-out. Each of the 13 sources was removed in turn and the eligible denominator and direction counts were recomputed. This identifies whether any single source drives the overall heterogeneity result.
(7) Fixed-station subset. Taiwan ODB gridded CTD cells were excluded, retaining only fixed monitoring stations. This tests whether gridded coastal products distort the direction pattern seen among traditional estuarine sensors.
(8) Direct-salinity-only subset. EC-derived salinity equivalents from USGS NWIS and GEMStat sources were excluded, retaining only records reported directly as salinity. This tests whether the conductivity-to-salinity conversion introduces systematic bias into trend counts.

Parallel all-observation trend fields were also generated to test whether flagged observations changed broad trend classes relative to the quality-control-pass primary screen.

2.5. Dataset characterization

The harmonized dataset contains 95,008,350 salinity records from 5383 source-aware station entries and 13 sources (Fig. 1). The records span 1963-2026 and include NERRS SWMP/CDMO (National Estuarine Research Reserve System Centralized Data Management Office, 2026), NOAA CO-OPS (NOAA Center for Operational Oceanographic Products and Services, 2026), USGS NWIS (U.S. Geological Survey, 2026), Ifremer QUADRIGE/REPHY (Ifremer, 2026), Chesapeake Bay Program tidal water-quality records (Chesapeake Bay Program, 2026), Delaware River Basin Commission water-quality records (Delaware River Basin Commission, 2026), UK Environment Agency water-quality archives (Environment Agency, 2026), RWS Waterinfo (Rijkswaterstaat, 2026), NSW EPA SEED (NSW Environment Protection Authority, 2026), selected PANGAEA estuary datasets (PANGAEA Data Publisher for Earth & Environmental Science, 2026), Taiwan Ocean Data Bank gridded CTD output (Taiwan Ocean Data Bank, 2026), and GEMStat-derived salinity records for Mexico and Argentina (GEMStat, 2026b; GEMStat, 2026a). GEMStat India records were excluded from the estuarine analysis after the reproducible coastline-distance screen and coordinate audit confirmed that the stations meeting the EC conductivity threshold were located on the Ganges-Yamuna plain (latitudes 8-29 N, longitudes 73-85 E), far from tidal influence; their elevated conductivity reflects irrigation drainage and industrial loads rather than marine input. Among the 436 trend-eligible source-aware station series, record spans ranged from 10 to 63 years (median 22 years, interquartile range 15-29 years). NERRS SWMP fixed-station series had the longest records (median 26 years), while Ifremer, DRBC, and NSW EPA series had median spans of 17-20 years. Most series (321 of 436, 73.6%) spanned 15 or more years, providing some separation from dominant decadal climate oscillations, though records shorter than 30 years remain susceptible to aliasing by ENSO and PDO variability.

Record volume is dominated by high-frequency North American monitoring: NERRS SWMP alone contributes 82,451,362 records, representing 86.8% of all observations. Station inventory is broadened by Ifremer, UK EA, CBP, Taiwan ODB and GEMStat, but many of those additional source-aware station entries are short, sparse, gridded, inland, or quality-control-sensitive compared with long fixed-station estuarine records. The map in Fig. 1 and Table 1 summarize source coverage and coordinate availability.

Quality control was designed to be auditable rather than destructive. All 95,008,350 records were retained in the canonical dataset while observation-level and metadata quality fields were added. In total, 11,212,494 records, or 11.80%, were flagged or routed for review. The most important high-flag cases are interpretable rather than simple failures: CBP freshwater-dominated grab-sample stations trigger flatline rules, GEMStat conductivity conversions include hypersaline and campaign-discontinuous records, and RWS stations include long stable salinity sequences. These flags define the primary analysis subset, but the flagged observations remain available for sensitivity analysis.

3. Results

3.1. Primary trend screen

Monthly station means were computed from quality-control-pass observations. A source-aware station series was eligible for the primary trend screen when it spanned at least 10 years and contained at least 60 observed monthly means. This produced 238,948 station-month rows and 436 trend-eligible source-aware station series.

The trend screen reveals heterogeneous station-level behavior across sources (Fig. 2). Nominal Kendall/Sen estimates classified 99 series as increasing, 76 as decreasing, and 261 as not significant at $p < 0.05$. Of

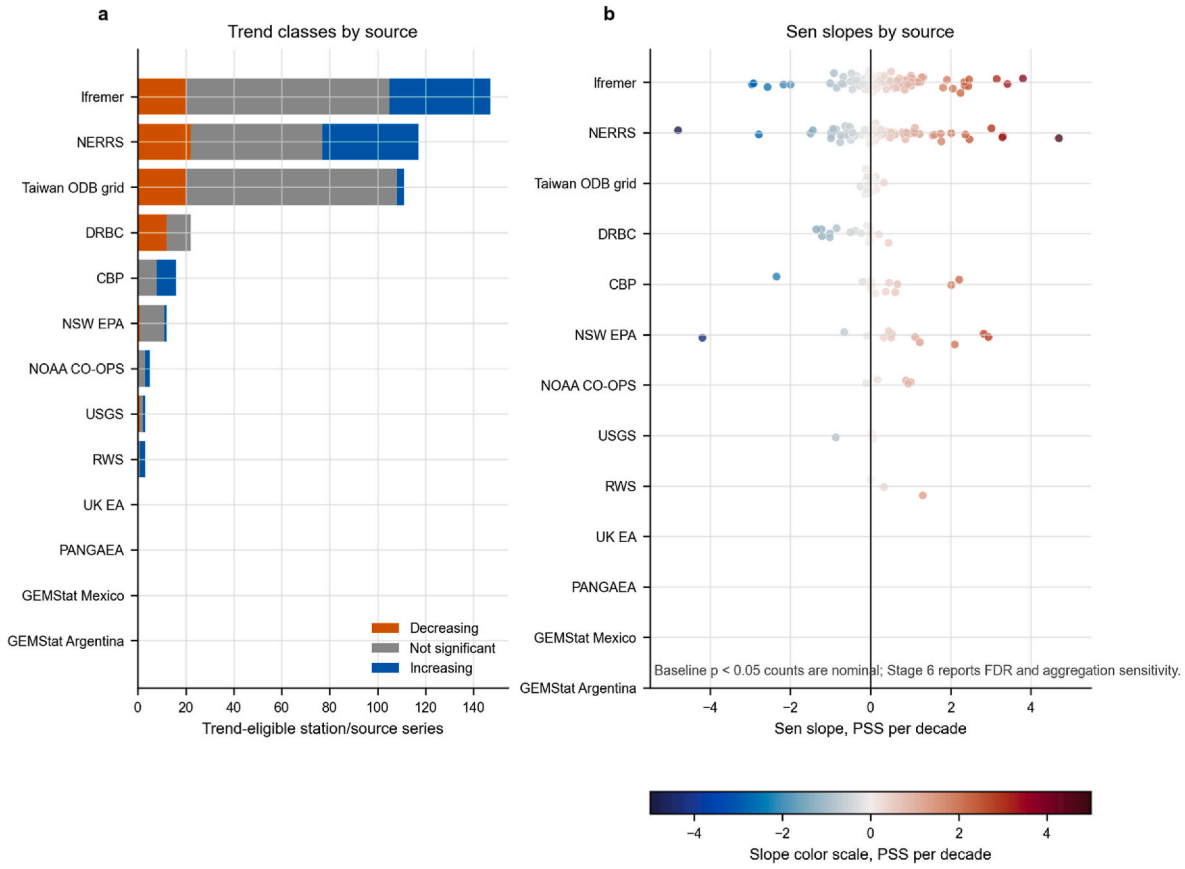


Fig. 2. Descriptive station-level trend screen. Baseline trend classes and Sen slopes by source for 436 source-aware station series meeting the primary eligibility threshold (10-year minimum span, 60 observed monthly means). Trend counts are nominal and should be interpreted alongside the sensitivity tests shown in Fig. 4.

the 175 nominally significant series, 56.6% were increasing, indicating a slight directional lean toward increasing salinity among detected trends,

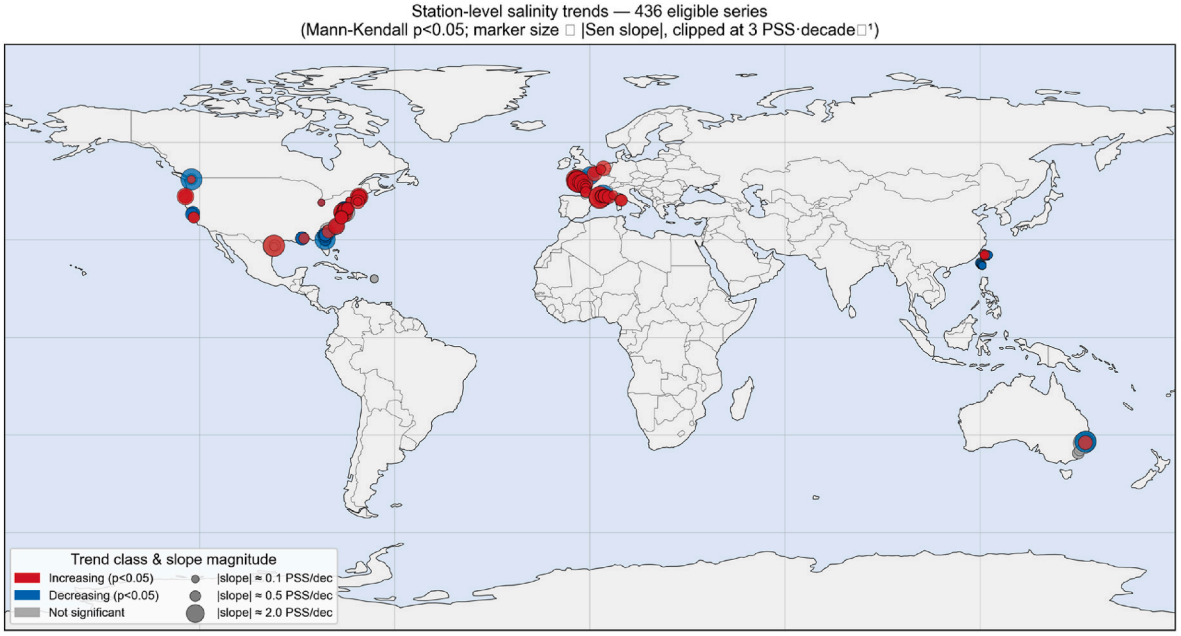


Fig. 3. Spatial distribution of station-level salinity trends. (a) Global overview: each point is one trend-eligible source-aware station series; color indicates trend class (red = nominally increasing at $p < 0.05$, blue = nominally decreasing, gray = not significant); marker area is proportional to absolute Sen slope (practical-salinity units per decade), clipped at 3 psμ/decade. (b) Regional detail panels for North America, Europe, East Asia (Taiwan ODB), and Australia (NSW EPA) are provided in Supplementary Fig. S1 to resolve overlapping markers on the global map. The map shows station-level screening results only and does not constitute a representative sample of all estuaries.

but not a uniform network-wide signal. These nominal values are descriptive screening results rather than the primary inferential endpoint. Increasing and decreasing trends coexist across all major sources, and the number of trend-eligible series differs sharply by monitoring program. Ifremer QUADRIGE/REPHY and NERRS SWMP dominate fixed-station trend evidence with 42 and 40 nominally increasing series respectively, Taiwan ODB contributes 111 gridded coastal context cells, and CBP adds many station entries but only 16 primary trend-eligible series after quality-control filtering. The DRBC network stands out as the most directionally coherent source: all 12 of its statistically significant series are decreasing, with zero increasing among 22 eligible series (binomial $p < 0.001$ for a decreasing majority), representing a regionally consistent freshening signal in the tidal Delaware system.

When mapped spatially, the direction and magnitude of these trends reveal distinct regional clusters rather than uniform salinization or freshening across the compiled archives (Fig. 3). Nominally increasing salinity trends are concentrated along the North American East Coast and Gulf of Mexico (NERRS SWMP stations) and the French Atlantic coast (Ifremer QUADRIGE/REPHY), consistent with long-term high-frequency monitoring in those regions. Decreasing trends are prevalent in the Delaware River system (DRBC; 12 of 12 significant series decreasing), in a subset of Ifremer stations in the French Atlantic, and across the Taiwan ODB gridded coastal zone (20 of 23 significant Taiwan series decreasing). The Sen slopes highlight localized hotspots of rapid change: NOAA CO-OPS harbor-entrance gauges exhibit the largest absolute slopes (median 0.88 practical-salinity units per decade), likely reflecting siting at high-mixing harbor entrances rather than representing broad estuarine trends. Marker sizes in Fig. 2 are proportional to absolute Sen slope magnitude, clipped at 3 practical-salinity units per decade to prevent a few outliers from dominating the visual scale.

Several sources mainly add inventory breadth rather than long-record trend evidence under the current rule. UK EA, PANGAEA, GEMStat Mexico and GEMStat Argentina contribute zero primary trend-eligible series. GEMStat India was excluded entirely from the analysis dataset after spatial auditing demonstrated that the EC-selected stations were located on the Ganges-Yamuna floodplain rather than in tidal estuaries (Section 2.1).

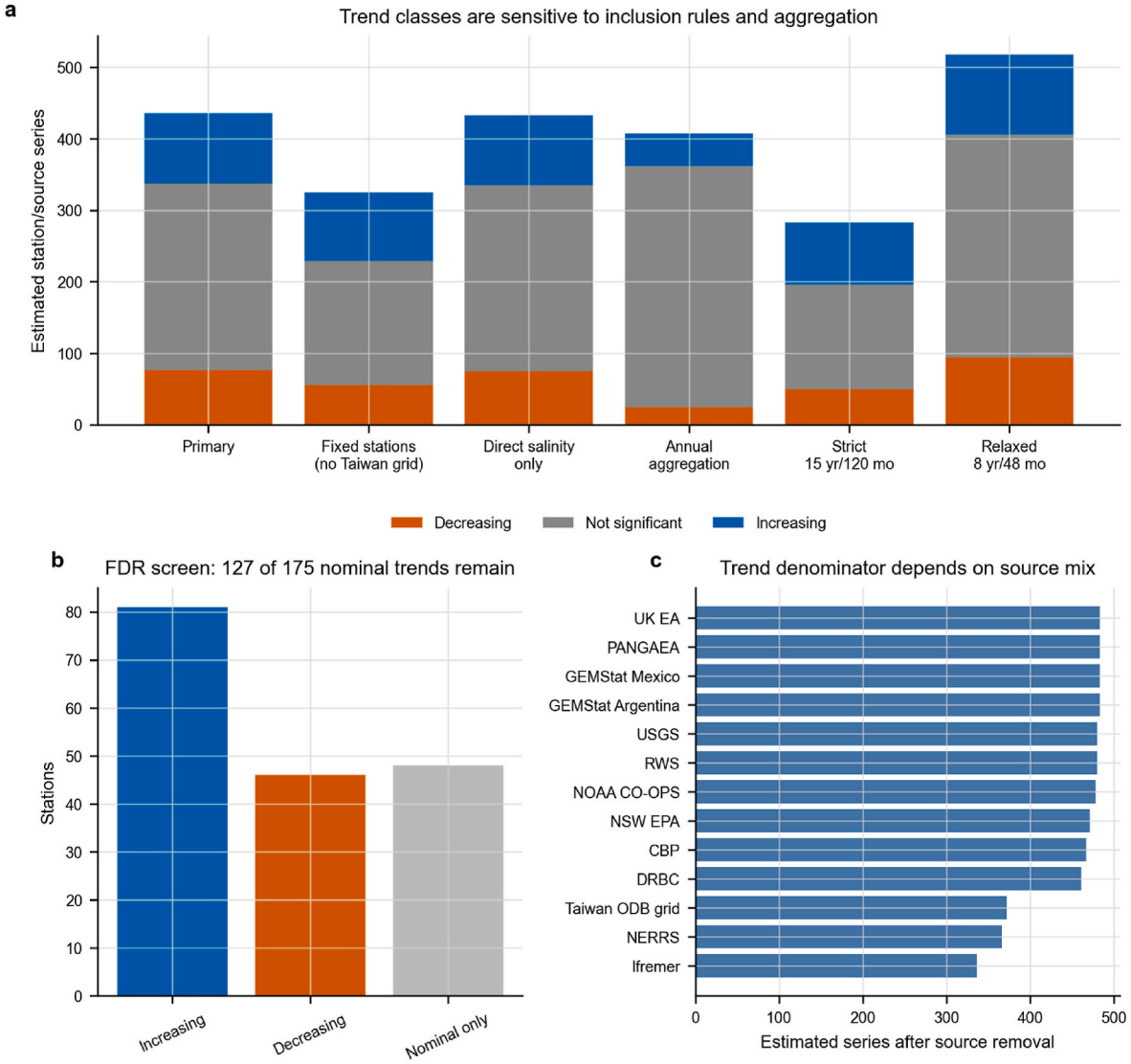


Fig. 4. Sensitivity of trend interpretation. (a) Stacked bars showing trend class counts under the primary eligibility rule, fixed-station-only subset excluding Taiwan ODB gridded cells, direct-salinity-only subset excluding EC-derived salinity equivalents, annual aggregation, a stricter 15-year/120-month rule, and a relaxed 8-year/48-month rule. (b) Benjamini-Hochberg false-discovery-rate (FDR) screen at $q < 0.05$: counts of series retained as significant versus the nominal manuscript baseline. (c) Source leave-one-out: estimated trend-eligible denominator after removing each source in turn. Together these checks support a qualified station-level trend screen rather than a broad pooled trend claim.

3.2. Sensitivity and robustness

The main trend conclusion is robust only at the level of heterogeneity. Exact counts of significant trends are sensitive to statistical and domain choices (Fig. 4). Excluding the Taiwan ODB gridded CTD source leaves a fixed-station subset of 325 estimated series, with 96 increasing, 56 decreasing and 173 not significant, confirming that the broad heterogeneity pattern is not an artifact of including gridded coastal cells alongside fixed estuarine sensors. Excluding EC-derived salinity equivalents and retaining only direct salinity records left 433 estimated series, with 98 increasing, 75 decreasing and 260 not significant, indicating that the overall trend-class distribution is not driven by conductivity conversion.

Multiple-testing and temporal aggregation further narrow the interpretation. A Benjamini-Hochberg false-discovery-rate screen at $q < 0.05$ retained 127 significant trends from the 436-series manuscript denominator, with 81 increasing and 46 decreasing. Annual aggregation produced 445 eligible annual series under the corresponding annual eligibility screen, with 58 increasing, 28 decreasing and 359 not significant. A stricter 15-year and 120-month inclusion rule retained 316 estimated series, whereas a relaxed 8-year and 48-month rule increased the count to 568. Serial dependence is also material: 279 of 436 trend-eligible series had absolute lag-one autocorrelation greater than 0.3, indicating that nominal significance counts should not be interpreted as independent detections. Applying TFPW across all 436 series shifted the counts to 116 increasing, 89 decreasing and 231 not significant (Fig. 6). The additional detections are concentrated in NERRS SWMP (nominal: 40 increasing, 22 decreasing; TFPW: 50 increasing, 26 decreasing) and Ifremer QUADRIGE/REPHY (nominal: 42 increasing, 20 decreasing; TFPW: 46 increasing, 21 decreasing) the two high-frequency fixed-sensor networks where tidal mixing generates the strongest positive serial correlation. Because TFPW can inflate Type I error under positive autocorrelation, these corrected counts are treated as sensitivity diagnostics rather than stronger evidence of additional true trends.

Quality-control sensitivity was comparatively stable but still important. Of the 436 trend-eligible series, 404 retained the same broad class when the all-observation series was used instead of the quality-control-pass series, while 32 changed class. These changed classes are concentrated on high-flag sources and should be reviewed before station-level results are used for local management interpretation.

3.3. Gridded versus fixed-station monitoring

The inclusion of gridded coastal products influences the distribution of apparent trend magnitudes. Taiwan ODB gridded CTD cells are therefore treated as coastal context rather than part of the fixed-station primary interpretation. They had a median absolute Sen slope of 0.047 practical-salinity units per decade (interquartile range 0.050), compared with 0.336 practical-salinity units per decade (IQR 0.773) for high-frequency fixed estuarine sensors from NERRS and Ifremer (Mann-Whitney $U = 5166$, $p < 0.001$; Fig. 5). This 7-fold difference suggests that gridded interpolation smooths out the localized tidal and discharge dynamics captured by fixed shoreline infrastructure and that gridded CTD products should be treated as spatial context rather than equivalents of fixed estuarine monitoring stations in trend analyses. A formal spatial autocorrelation test of trend sign directions (Moran's I) would further characterize whether regional neighbor stations share trend class beyond chance; this is identified as a priority for follow-up analysis.

4. Discussion

4.1. Heterogeneity as the primary result

The strongest result from this synthesis is the absence of a single coherent direction of salinity change across public estuarine and coastal monitoring archives. Increasing, decreasing, and not significant trend

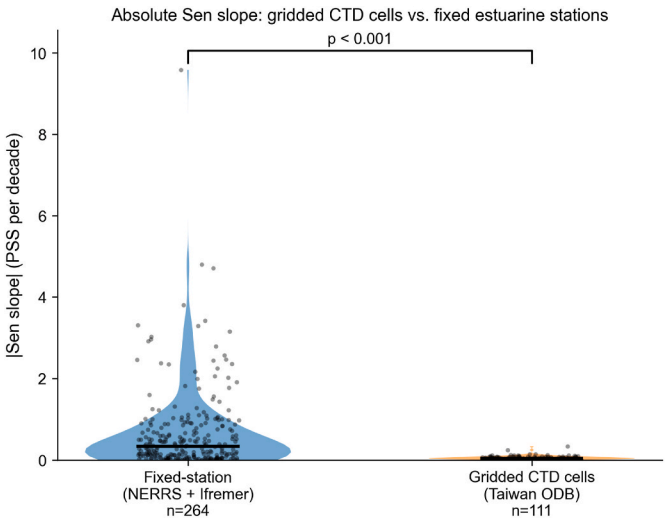


Fig. 5. Absolute Sen slope comparison: gridded CTD cells vs.fixed estuarine stations. Gridded Violin plots show the distribution of absolute Sen slopes (practical-salinity units per decade) for Taiwan ODB gridded cells (n = 111) and NERRS/Ifremer fixed estuarine stations (n = 264). Individual points are jittered. Medians shown as horizontal bars. Mann-Whitney $U = 5166$, $p < 0.001$.

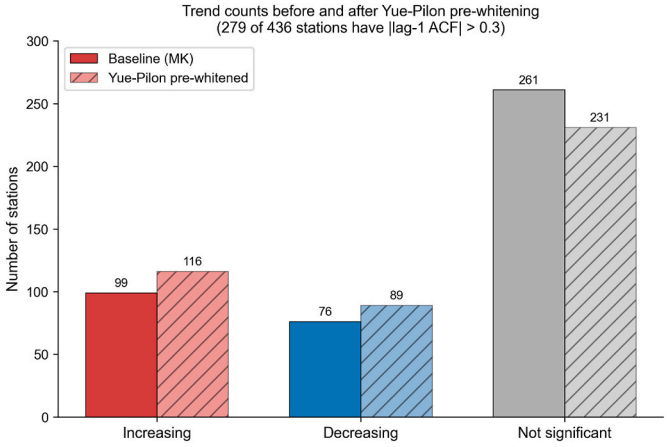


Fig. 6. Trend counts before and after trend-free pre-whitening. Grouped bar chart showing the number of source-aware station series classified as increasing ($p < 0.05$), decreasing ($p < 0.05$) or not significant before (nominal Kendall) and after TFPW correction for serial dependence. The corrected counts are reported as a sensitivity diagnostic because TFPW has known limitations under positive autocorrelation.

classes all occur among the 436 eligible source-aware station series, and the exact counts shift under conservative sensitivity tests including multiple-testing control, annual aggregation, and autocorrelation correction. This is not a null result: the heterogeneity itself is informative. It shows that broad salinization narratives derived from pooled monitoring archives are not supported by the current evidence base, and that regional signals must be interpreted separately by source, geography, and record length.

4.2. Regional signals and candidate drivers

Although coherent change across all compiled records is absent, the harmonized archive enables the identification of regional signals that warrant attribution-focused follow-up. The Delaware River Basin Commission network provides the clearest example. All 12 statistically significant DRBC trend series are decreasing (zero increasing among 22 eligible series), with a median Sen slope of −0.031 practical-salinity

units per decade over the 2000-2021 observation window. This regional freshening signal in the tidal Delaware system stands in apparent contrast to the salinization signal reported by Ross et al. (2015) for the same system in an earlier period, suggesting a testable hypothesis that trend direction shifted during the intervening decades. Candidate mechanisms include changing freshwater discharge linked to altered upstream precipitation and reservoir management, sea-level-driven changes in tidal prism, and shifts in salinity stratification.

Along the North American East Coast and Gulf of Mexico, 40 NERRS SWMP series show nominally increasing annual salinity trends. Hypothesized drivers include sea-level acceleration increasing tidal and storm-surge penetration, land-use change in tributary watersheds reducing freshwater dilution, and reduced riverine discharge during dry periods (Kaushal et al., 2018; Li et al., 2025). Along the French Atlantic coast, 42 of the 42 nominally increasing Ifremer QUADRIGE/REPHY series also show increasing trends, consistent with reduced Atlantic freshwater inputs and long-term circulation changes. The Taiwan ODB coastal cells show predominantly decreasing salinity (20 of 23 significant series decreasing), which may reflect increasing freshwater discharge from Taiwan's steep, typhoon-influenced catchments or artefacts of the gridded interpolation scheme. These regional signals are the most scientifically actionable output of this study, and each represents a hypothesis-testing opportunity for targeted hydrodynamic or statistical attribution work.

4.3. Internal climate variability and record-length limitations

Interpreting decadal salinity trends in records spanning 10-30 years requires explicit acknowledgment of internal climate variability. The El Nino-Southern Oscillation (ENSO), Pacific Decadal Oscillation (PDO), North Atlantic Oscillation (NAO), and Atlantic Multidecadal Oscillation (AMO) all modulate precipitation, freshwater discharge, and coastal sea level over periods of 3-70 years (Milly et al., 2008). A monitoring record of 20 years may capture one to two full cycles of ENSO and PDO, which are sufficient to generate apparent positive or negative salinity trends that do not reflect long-term directional change. The majority of fixed-station series in this study (median 22 years) fall within this range. This limitation is not unique to this synthesis: it applies to any trend screen of estuarine monitoring archives available today. Reporting individual record lengths (Section 2.5) and applying multi-test sensitivity (Section 3.2) are the primary mitigations available within the current data constraints. Extending records or pooling cross-archive station neighborhoods with overlapping observation periods would reduce the aliasing risk in future updates.

4.4. Source dependence and monitoring design

Source dependence is itself a substantive finding. Long, high-frequency fixed-station programs (NERRS SWMP, Ifremer QUADRIGE/REPHY) dominate trend eligibility, whereas many geographically valuable archives do not yet contribute long primary trend series. This distinction matters for future synthesis work because adding more observations is not equivalent to adding more comparable long records. CBP, UK EA, GEMStat, PANGAEA, and Taiwan ODB each improve the resource in different ways, but their records represent fundamentally different sampling designs. The GEMStat India case illustrates why spatial auditing must be part of estuarine salinity synthesis: some conductivity-derived salinity records pass a conductivity threshold but are located far from tidal influence. Retaining such records in the archive with appropriate provenance flags, while excluding them from the estuarine trend analysis, is preferable to either silent deletion or uncritical inclusion.

4.5. Limitations

Several limitations constrain the scope of inference. First, quality-control decisions affect trend classes: of the 436 trend-eligible series, 32 (7.3%) changed broad direction class when flagged observations were included alongside quality-control-pass observations. These changed classes are concentrated in high-flag sources (CBP, GEMStat) and should be reviewed before station-level results are used for local management interpretation. Second, the flatline QC threshold (96 consecutive identical values) may incorrectly flag genuine low-variability sequences in well-mixed, macro-tidal environments where salinity changes slowly over multiple days. Third, the EC-to-salinity linearization used for USGS NWIS and GEMStat conductance records is least defensible in freshwater-influenced stations with non-marine ionic composition (Kaushal et al., 2018; Fofonoff and Millard, 1983); the direct-salinity sensitivity test confirms that excluding these records does not materially change the direction count distribution (98 increasing, 75 decreasing, 260 not significant versus the primary 99/76/261). Fourth, the discharge-change QC threshold applied to sub-hourly jump detection (15 practical-salinity units/hr) may incorrectly flag rapid desalination events in systems with flashy freshwater discharge; future updates should condition the jump flag on co-located discharge data where available. Fifth, NERRS stations are not randomly sited: they reflect reserve monitoring objectives and may oversample ecologically sensitive or management-relevant environments, introducing geographic selection bias into the network-level direction counts.

4.6. Implications for monitoring synthesis and future work

The analysis identifies where the next scientific advance should occur. Stations with long records, stable quality-control sensitivity, and strong nominal slopes - primarily in NERRS SWMP, Ifremer QUADRIGE/REPHY, DRBC, and NOAA CO-OPS - are candidates for driver attribution using river discharge, precipitation, sea level, tidal range, bathymetry, dredging, withdrawals, and station metadata. Conversely, sources that add many sites but few eligible trend series (UK EA, PANGAEA, GEMStat, Taiwan ODB for fixed-station purposes) show where archive completion, metadata repair, or targeted retrieval would improve the study more than additional statistical refinement. A formal spatial autocorrelation analysis (Moran's I) of trend sign directions would further characterize whether regional neighbor stations share trend class beyond chance; this is identified as a priority for follow-up.

5. Conclusions

This study assembled a large, source-aware salinity resource for estuarine and coastal monitoring, comprising 95,008,350 harmonized records from 5383 source-aware station entries and 13 public sources. The resulting evidence base is substantial but spatially and institutionally unbalanced. Its scientific value is threefold: it establishes an auditable observational baseline that can anchor validation of hydrodynamic salt-intrusion projections (Lee et al., 2024, 2025; Li et al., 2025); it makes source dominance, coverage gaps, and quality-control decisions visible and reproducible; and it identifies specific regional signals - particularly the tidal Delaware freshening trend and the North American East Coast and French Atlantic salinization clusters - that are coherent enough to support targeted attribution follow-up.

Across trend-eligible source-aware station series, no single direction of salinity change emerged from the compiled public archives. Instead, increasing, decreasing and not significant trends co-occurred, and the exact counts changed under reasonable sensitivity tests. This pattern argues for caution in broad salinization narratives based only on pooled public monitoring archives, while also showing that many local and regional salinity changes are detectable and worth attribution-focused follow-up.

Future progress will depend less on another omnibus trend test than on completing underrepresented regional archives (especially UK EA, PANGAEA, and Southern Hemisphere sources), improving station

metadata, distinguishing fixed stations from gridded or campaign products, and linking robust trend candidates to hydrologic and geomorphic drivers including river discharge, sea level, tidal range, and land-use change (Kaushal et al., 2018; Beemster et al., 2026; Lee et al., 2025). Extending records beyond 30 years in the highest-priority estuaries would substantially reduce aliasing by internal climate variability (Milly et al., 2008). The workflow developed here provides an auditable starting point for that next step.

Funding

This research received no external funding.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The author thanks the data providers whose public archives make this synthesis possible: NOAA/NERRS CDMO, NOAA CO-OPS, USGS NWIS, Ifremer, the Chesapeake Bay Program, the Delaware River Basin Commission, the UK Environment Agency, Rijkswaterstaat (RWS Waterinfo), NSW EPA, PANGAEA, the Taiwan Ocean Data Bank, and GEMStat.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ecss.2026.110103.

Data availability

Data will be made available on request.

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